Expressive and Efficient TTS via Synthetic Data

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Abstract

Recent advances in phoneme-based text-to-speech (TTS) systems have achieved impressive speech quality, compact model sizes, and fast inference speeds. However, these models still lack fine-grained control over expressive and paralinguistic aspects of speech that are essential for natural human-like synthesis. In this work, we propose a simple way to transfer such expressive control into lightweight models by distilling it from a larger TTS system trained on proprietary data. Instead of using that data directly, we generate a synthetic distillation dataset that reflects diverse emotional and paralinguistic variations. A phoneme-based student model is then trained on these expressive samples using a masking loss design that disentangles control features from speech content. Our results show that the distilled model attains expressive control and intelligibility comparable to much larger systems while being significantly faster and smaller. This suggests that expressive capability can be treated as a transferable property rather than a feature limited to large proprietary models. Our code is available at `https://github.com/BarryFutureman/NeuTTS-Express`.

1 Introduction

Text-to-speech (TTS) synthesis is the task of converting text into natural-sounding speech [9]. In the current times, there has been significant progress in integrating controllable emotional expressivity into TTS, bringing natural, nuanced performance into the speaking models. To cite an example, EMOQ-TTS [13] and QI-TTS [29] both achieved significant improvement in the MOS scores compared to previous studies.

Despite the remarkable breakthroughs with large generative TTS models, systems constrained to a small model with streamable operation remains a critical challenge. To address this problem, there are several efforts being made. For instance [20], one approach is to use a pipelines based on phonemes (basic sound units in linguistics) rather than direct text-to-speech. Early neural approaches such as FastSpeech2 [23] and VITS [16] used a phoneme-based representation for faster, more robust, and higher-quality synthesis. However, even with these advances, such phoneme-based systems predominantely optimise for intelligibility and fluency rather than emotional or stylistic variation, leading to speech that, while realistic, often lacks the subtle prosodic diversity characteristics of human expression.

Even though these advances improved fluency and speed [26], the models lack fine-grained expressive control, the ability to manipulate emotion, prosody, and other paralinguistic cues, which remains an open challenge in neural TTS because of several hurdles [31]. In addition, enabling fine-grained emotional control on these models still often depends on large-scale of data. In practical deployment scenarios, access to such a vast amount of data is not always feasible.

Researchers have introduced various mechanisms for incorporating prosody and style modeling. Notable examples include Global Style Tokens (GSTs) [34], prosody reference encoders [28], which allow unsupervised style control through learned embeddings. Other methods have leveraged disentangled latent representations for emotion transfer and expressive control [17]. However, most expressive TTS rely on small, emotion labelled datasets, such as IEMOCAP [2] and RAVDESS [19], which provide limited speaker and linguistic diversity. Training end-to-end expressive models on such constrained data often leads to overfitting or degraded base model quality. Consequently, large-scale expressive control remains largely confined to proprietary models trained on extensive internal datasets, creating a disparity between research and deployable expressive synthesis [3, 32].

In this work, we aim to bridge that gap by leveraging the capabilities of existing large expressive TTS models to distill expressivity into smaller open models. After reading about the success of knowledge distillation [11] and dataset distillation [33], we propose a novel framework in which a proprietary expressive model is used as a teacher to distill this capability into the student model. To achieve this, we will add tags to the sentences in a synthetic dataset with different sentences with emotions. This distillation is then used to train a lightweight, phoneme-based TTS model capable of controllable and expressive synthesis without direct access to large proprietary data. By decoupling the expressive modeling process from the original dataset, our approach enables open models to inherit expressive control while maintaining efficiency and accessibility.

Our study contributes to the ongoing effort to make expressive TTS both democratised and efficient. We demonstrate that synthetic data derived from expressive teacher models can transfer fine-grained control to smaller student architectures, improving prosodic richness and emotional variation without sacrificing intelligibility or speed. Through this work, we provide evidence that distillation-based expressivity transfer is a viable path towards open, scalable, and human-like TTS.

2 Background and Related Work

Human-like spoken dialog systems increasingly require not just intelligible speech but also expressive emotional speech which can convey emotion, empathy, paralinguistic cues that support engagement and natural interaction [1]. A major challenge in emotional text-to-speech (TTS) synthesis is the lack of sufficient good-quality annotated datasets. Consequently, many systems struggle with fine-grained emotion control and exhibit a strong entanglement between speaker identity and emotion embeddings, which hinders emotion transfer across speakers [1, 30].

To enhance expressive and emotional control, recent work has emphasised disentangled representations, enabling separation of speaker identity, emotion/style, and prosody. For example, multi-emotion, multi-speaker, multi-lingual systems such as Mels-TTS [5] adopt disentangled style tokens for speaker/style/emotion separation. Related to this, similar work DiEmo-TTS [4] explicitly pursues speaker-independent emotion embeddings via self-supervised distillation, improving cross-speaker emotion transfer.

Classical approaches include Global Style Token (GST) that embed style clusters [34], VAEs [16] or flow-based encoders for latent style/emotion modelling [31]. Contemporary systems aim for emotion intensity control, moving beyond categorical labels to continuous control parameters [10, 13, 31]. Furthermore, zero-shot emotion control has emerged with models that synthesise emotional speech for previously unseen speakers and unseen styles by leveraging pretrained emotion embeddings or reference speech. For example, ECE-TTS [18] achieves zero-shot emotion control through Emotion-Adaptive Spherical Vectors (EASV) derived from Valence-Arousal-Dominance (VAD) features, enabling fine-grained emotional intensity modulation.

Recent research has made substantial progress toward systems that enable emotional style control beyond training speakers or styles. Some models like the ECE-TTS [18] present zero-shot emotion-controllable TTS based on the F5-TTS architecture. It simplifies emotion modeling using pretrained emotion recognizers to extract VAD values and convert them into EASV representations for fine-grained control, eliminating the need for separate regression heads. Another model introduces hierarchical emotion distribution modeling that combines BERT semantic embeddings, supervised learning, and ranking functions to modulate emotion intensity and distribution within a single utterance, enabling multi-emotion control in one sentence [14]. There is also a diffusion-based emotional TTS model, EmoDiff [10], that uses soft-label guidance, inspired by classifier guidance, to allow continuous emotion intensity control. However, it remains computationally heavy and often depends on discrete labels or limited speaker data. In other works, GenerSpeech [12] explores style transfer for out-of-domain (OOD) generalization, providing a foundation for adaptive emotion transfer across diverse datasets. While Mels-TTS [5] demonstrates multi-speaker, multilingual, and multi-emotion synthesis using disentangled style tokens, further validating the role of representation disentanglement for flexible style modeling. These systems show progress but still often depend on discrete labels or limited speaker sets.

While prior studies have advanced emotional control, they reveal a persistent trade-off between expressivity and efficiency. Methods that achieve fine-grained control such as flow-matching based architectures [18] can introduce additional computational cost at inference. Similarly, training pipelines also become more complex, for example relying on external emotion predictors [14] or soft-label guidance [10] and self-supervised distillation for speaker-invariant emotion embeddings [4]. Moreover, control interfaces such as per-phoneme emotion vectors [14] provide high resolution but are less intuitive for end users because they require precise time alignment. To mitigate these challenges, recent research has explored efficient neural audio compression as a means of maintaining expressivity while improving inference speed and deployability. Neural Audio Codecs (NACs) [38] compress speech into discrete latents for efficient TTS. Unlike traditional vector quantization (VQ) and residual VQ, Finite Scalar Quantization (FSQ) encodes each latent dimension into finite scalar levels, improving codebook utilization, noise robustness, and compression [15]. In expressive TTS, FSQ enables real-time, low-latency emotional synthesis, balancing expressivity and efficiency [15].

Our work targets these limitations by starting from an efficient base model called NeuCodec, a fast phoneme-based pipeline and a neural audio codec augmented with FSQ quantization [15]. We introduce a simplified training scheme that adds expressive capability to this compact backbone and enables discrete, tag-based control, retaining controllable expressivity comparable to larger systems while keeping a lightweight, streamable runtime.

3 Data

We use EmoVoice-DB [37], an English emotional speech corpus synthesized with GPT-4o models. It contains 22,100 utterances (approximately 40.45 hours) rendered with five speaker timbres and balanced across seven emotions: angry, happy, sad, surprised, fearful, disgusted, and neutral. We draw 100 utterances per emotion for test and 50 per emotion for validation, leaving 21,050 utterances for training. This preserves both emotion balance and the diversity of text styles, including narration, dialogue, and descriptive lines. Each example is a single JSON record with fixed, explicit fields. For our model, we retain a unique key, the target text to be spoken, an emotion label, and a file path to a mono 16kHz waveform.

Our transformer decoder adopts standard teacher-forcing training, the model predicts each speech token y_t conditioned on previous tokens $y_{<t}$ and input text/control tags X, i.e., $\hat{y}_t = f_\theta(y_{<t}, X)$, where f_θ is the TTS model. If we decompose each speech token into speaker characteristics s_t and tone e_t, we have $\hat{y}_t \approx s_t + \tilde{e}_t(y_{<t})$, where $\tilde{e}_t(y_{<t})$ is inferred from the previous speech sequence. In this scenario, the tone loss $\mathcal{L}_{\text{tone}} = \sum_t \|e_t - \tilde{e}_t(y_{<t})\|$ is near zero, indicating that the model does not learn the mapping from the control tag to tone.

To enforce tone disentanglement, we introduce an anchor segment. Each training sample is formally represented as

$$X(context) = \underbrace{\texttt{<neutral> TEXT\_A}}_{\text{neutral\_anchor}}\ \underbrace{\texttt{<E> TEXT\_B}}_{\text{target\_emotion}}, \underbrace{\texttt{SPEECH\_A}}_{\text{neutral speech}}, \quad Y = \underbrace{\texttt{SPEECH\_B}}_{\text{target\_emotion\_speech}},$$

where E is the target emotion. The loss is computed only on the target emotion speech tokens as $\mathcal{L}_{\text{masked}} = \sum_{t \in \text{SPEECH\_B}} \ell(y_t, f_\theta(y_{<t}, X))$. Because the initial tokens of SPEECH_B are conditioned mostly on the anchor, the model cannot copy emotional tone from context and must rely on the control tag <E> to generate the correct tone, i.e., $\hat{y}_t \approx s_t + e_t(\texttt{<E>})$. This disentanglement approach preserves speaker characteristics for zero shot voice cloneing, enhances boundary sensitivity and controllability at inference.

Furthermore, we generate 33.59 hours of synthetic distillation data using *Maya1* [24], a 3-billion-parameter text-to-speech model designed for expressive voice generation with rich emotional nuance and precise voice controllability. In the first step, we use the *Qwen2.5-Instruct 7B* model [21] to produce a diverse set of prompts. (We chose to use this model due to its simplicity and stability for without reasoning). Given a random topic, the model generates sentences augmented with paralinguistic tags such as <laugh>, <cry>, or <sigh>. In the second step, we synthesize

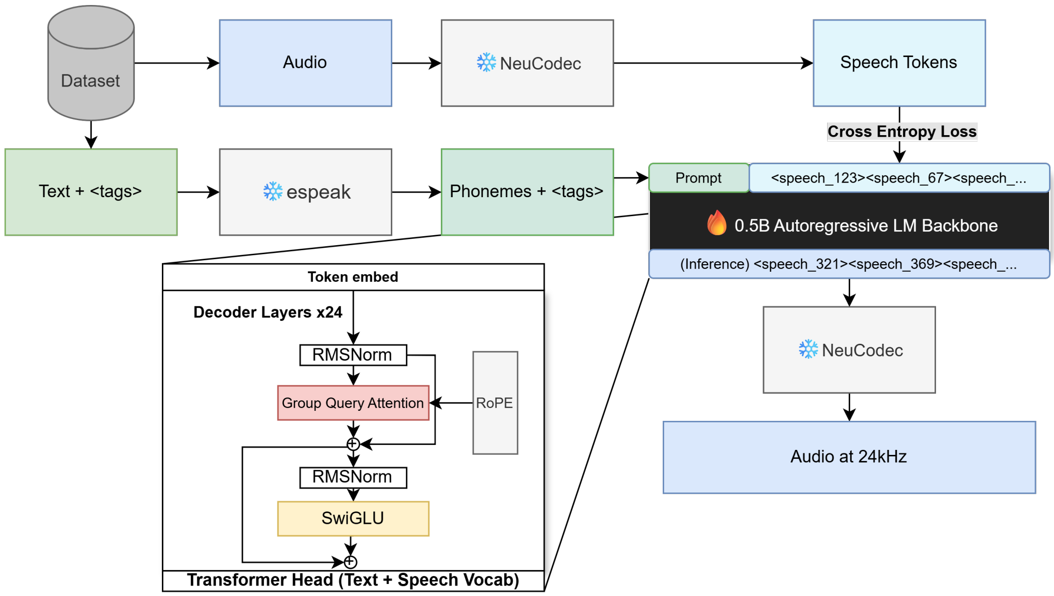

Figure 1: TTS System Architecture Overview. We show how an audio text pair is converted and placed to the single stream of a transformer, trained via cross entropy loss and generate speech tokens at inference. Bottom left: LM Transformer Decoder

corresponding audio samples using these generated prompts combined with randomized speaker attributes, including timbre, gender, accent, and age. We distribute the workload on 16 RTX4080 desktops. The resulting dataset captures a broad range of expressive and stylistic variations. We finetune our model on this distillation set to acquire the ability to reproduce paralinguistic cues and expressive nuances in generated speech.

The EmoVoice-DB dataset is released under MIT license while the Maya1 model is released under Apache 2.0 with no further restrictions. Thus, there is no legal implications from using these data to train our model. This combined corpus provides the necessary diversity and structure to support emotion-controllable, expressive speech synthesis.

4 Model Architecture

4.1 Architecture details

Our TTS system builds upon NeuTTS Air [15], which consists of two main components: a 0.5B parameter transformer initiazed from Qwen2.5 0.5B [36] and a neural audio codec. To generate speech from text, the input text is first converted to phonemes (linguitic units of sound) using *espeak-ng* [8], with stress markers and preserved punctuation, then tokenized and fed into the LM transformer. The language model generates audio tokens autoregressively, which are then decoded by the NeuCodec model [15] to produce high-fidelity audio. NeuCodec is an efficient neural codec that uses Finite Scalar Quantization (FSQ) to encode audio to discrete tokens at just 0.8 kbps, using both an audio encoder (BigCodec [35]) and a semantic encoder (Wav2Vec2-BERT [6]) to capture acoustic and linguistic information, and upsampling from 16kHz to 24kHz for natural-sounding speech.

To enable emotional and paralinguistic control, we include special tags such as `<giggle>`, `<sigh>`, or `<sad>` within the input text. During phonemization, these tags are preserved and passed through tokenization as text to guide expressive synthesis. We do this to prevent the model from confusing control tags with words that needs to be read out loud.

Figure 1 showcases an overview of the TTS system, while Figure 1 (bottom left) highlights details on the backbone decoder.

Specifically, the decoder backbone 1 is a 24-layer dense transformer. The vocabulary has 217,652 tokens, encompassing text tokens, 65,536 added speech codes, and special tokens.

4.2 Training Details

For each text-audio pair from our dataset, we tokenize them via the methods described above. The tokens are arranged in a single stream similar to standard language modeling. The prompt is placed at the front, followed by the speech tokens. We apply the loss only to the speech tokens. The model is trained with standard cross-entropy loss on the target speech token predictions; we mask the loss on non-target positions. During training, only the language model based decoder is active while other aspects of the model remain frozen. We full rank fine-tune the pretrained NeuTTS Air model using the following hyperparameters on a single RTX4080:

- **Precision**: bfloat16
- **Attention**: Flash Attention 2 [7]
- **Max sequence length**: 768 tokens
- **Learning rate**: 4×10^{-5} with cosine annealing schedule
- **Batch size**: 4
- **Training epochs**: 4 (EmoVoice-DB) + 6 (Full Dataset)
- **Optimizer**: AdamW

5 Results

We evaluate our model against the 3B Maya1 baseline across three dimensions: objective metrics (intelligibility and speed), batch-level stability, and human perceptual evaluation. Together, these results assess both technical performance and perceived expressive quality. We chose Maya1 [24] as a baseline because it was released recently (November 2025), and it is the current state-of-the-art in controllable, codec-based TTS models.

5.1 Objective Evaluation

We evaluate 100 test sentences using two metrics. First, Word Error Rate (WER), computed via OpenAI Whisper (base), measures intelligibility through transcription accuracy. Second, Real-Time Factor (RTF) measures generation time relative to audio duration (RTF < 1.0 indicates faster-than-real-time synthesis). Table 1 summarizes the results.

Table 1: Objective comparison between our model and Maya1.

Model	**Condition**	**WER**	**RTF**	**Speedup**
Maya1 (Baseline)	Clean	8.27%	1.10	1.0×
Ours	Clean	2.89%	0.51	2.16×
Ours	Emotion	8.73%	0.51	2.16×

Despite using only 0.5B parameters (1/6th of Maya1), our model achieves a threefold WER reduction (2.89% vs. 8.27%) in clean speech and generates audio 2.16× faster, enabling real-time use on consumer hardware. Adding emotion tags slightly increases WER (8.73%), but performance remains comparable to Maya1, indicating expressive control can be added with negligible quality loss and zero computational overhead.

5.2 Batch-Level Stability

Aggregate scores can mask inconsistent behavior, so we divide the 100-sentence test set into 10 batches of 10 sentences to measure stability. Table 2 reports the WER distribution across batches.

Table 2: Per-batch WER (%) stability across 100 test sentences.

Metric	Ours (Clean)	Maya1 (Clean)	Ours (Emotion)
Mean WER	2.89%	8.27%	8.73%
Std. Dev.	**2.10%**	7.91%	5.16%

Our model exhibits nearly 4× lower variance than Maya1 (2.10% vs. 7.91%), indicating predictable performance across diverse inputs. Maya1 alternates between perfect and highly erroneous generations, showing instability that would hinder production deployment. Our model maintains consistent quality even when generating emotional speech, with standard deviation increasing only to 5.16%.

5.3 Mean Opinion Score Evaluation

To assess perceptual quality and emotional expressiveness, we conducted a listening study with 11 participants. Each participant rated speech samples across 7 emotions (angry, happy, sad, surprised, fearful, disgusted, neutral) on two 5-point Likert scales: **naturalness** (1 = very unnatural, 5 = very natural) and **emotion match** (1 = does not match, 5 = perfect match). We generated samples from both our model and Maya1 using identical text prompts and emotion tags. Results are summarized in Table 3.

Table 3: Human evaluation Mean Opinion Scores.

Model	Naturalness	Emotion Match
Maya1 (Baseline)	3.17	2.82
Ours	**3.99**	**3.91**

Our model significantly outperforms Maya1 on both metrics, achieving higher naturalness (3.99 vs. 3.17) and better emotion match (3.91 vs. 2.82). These results demonstrate that our lightweight model not only generates more natural-sounding speech but also captures emotional nuances more effectively than the 6× larger baseline.

6 Discussion

Typical TTS systems trade accuracy and quality for speed or model size. Our results show that careful architecture and distillation outperform this assumption. Our 0.5B model is both *faster*, *more accurate* and matches larger models in capabilities. Emotional control adds no computational cost, since it is encoded within learned representations rather than architectural modifications. Thus, our results support the core claim that our augumented synthetic data enables compact, accessible expressive TTS models without requiring large proprietary datasets. The success of our model can be attributed to three major points discussed below.

NeuCodec vs SNAC. Our model uses NeuCodec while the baseline Maya1 uses SNAC [27]. Although SNAC demonstrates superior reconstruction quality (Table 4), it employs complex token structure and density that places significant stress on the language model backbone, necessitating Maya1's larger model size. NeuCodec is more LLM-friendly and closer to language space, enabling our compact 0.5B decoder. The inference slowdown from NeuCodec is less than 0.1 seconds, which is negligible in practice. This architectural choice demonstrates that codec selection involves trade-offs beyond reconstruction quality alone-inference efficiency and model compatibility are equally critical for deployment.

Phoneme-based vs Text-based. Our model converts text into phonemes while the baseline directly takes text as input. Grounding the model to phonemes ensures correct pronunciation regardless of data distribution, as evidenced by our superior WER stability across batches. Additionally, phoneme representation handles part of the text-to-acoustic mapping, reducing the learning burden on the decoder and enabling our much smaller architecture.

Table 4: SNAC (2024) vs NeuCodec (2025).

Feature	**SNAC**	NeuCodec (distilled encoder)
Bitrate	0.98 kbps	**0.80 kbps**
MUSHRA (clean channel)	**86–89**	78–82
STOI	**0.932**	0.905–0.918
PESQ (NB)	**2.18**	2.04–2.11
10% random bit flips	unusable	**STOI 0.87, intelligible**
Token structure	3 streams	**single flat sequence**
Encoder params (distilled)	**6.7 M**	57 M (36M acoustic + 21M semantic)
Semantic grounding	acoustic only	**DistilHuBERT frontend**

Synthetic Data vs Labeled Real Data. Our fully synthetic approach offers several advantages. Real data is often noisy and high-variance, and manual labeling introduces potential hallucinations depending on accuracy, while synthetic data can be generated rapidly at scale. Our batch-level stability results (Table 2) suggest synthetic data produces consistent training signals. Further, it is difficult to disentangle semantics with emotions, but synthetic data mitigates this issue. The primary limitation is that student quality is upper-bounded by the teacher model. Future work may explore fine-tuning on high-quality real data or applying preference optimization [22] to exceed teacher performance in specific aspects.

However, although these results are promising, our current model exhibits several issues worth examining. First, We observe that the model produces slower speech when conditioned on certain emotion tags. This behavior is mainly influenced by the EmoVoice-DB data. Further, given emotion tags increases word error rate as discussed earlier. When pushing for more stylization the model also tend to be more unstable.Secondly, paralinguistic tags can exhibit inconsistent realization, sometimes producing extended pauses or muted expressions. This can be noticed because of noise introduced by the synthetic data pipeline, where the generating model itself produces imperfect paralinguistic cues. Another reason for the paraliguistic accuracy can be the frozen NeuCodec, which was trained primarily on clean speech and may not faithfully encode non-speech vocalizations. Lastly, insufficient training to fully capture the distribution of these relatively rare events could also be a factor.

7 Limitations

In our experiments, we were constrained quite heavily by computation. All training had to run on a single RTX 4080 (16GB VRAM), which meant our batch size had to stay at four, and we did not really have room for any substantial hyperparameter exploration. This shows up clearly in the training logs: the loss curve fluctuates at a fairly high frequency, which is usually a sign that the gradient estimates are pretty noisy. The model still managed to adapt reasonably well within about five epochs, but it is clear the optimization process was not operating under ideal conditions.

Another limitation comes from the very low bitrate (0.8 kbps) of the neural codec. The NeuCodec tokenizer was trained on clean speech, so it struggles when we move into more irregular emotional speech, like the vocal fry that often appears in sadness or the breathy quality of whispered tones. These non-tonal components do not quantize well at such a low bitrate, and the resulting artifacts can noticeably hurt ASR performance, sometimes inflating the WER even if the overall prosody still sounds acceptable.

We also run into structural issues stemming from our reliance on synthetic data and discrete emotion tags. The student model tends to inherit some of the quirks and biases of the teacher model, and because our emotional labels are mutually exclusive (e.g., <sad>, <laugh>), the system cannot express blended or more nuanced emotional states. Cases like “sorrowful laughter” or “sarcastic joy” simply can not be represented without moving to some kind of continuous emotional embedding space.

Finally, our evaluation scope is limited. Whisper-based WER gives us an objective measure of intelligibility, but it tells us nothing about expressiveness. Our human evaluation involved only a small number of participants and isolated sentences, which naturally restricts the conclusions we can draw.

8 Ethical Considerations

While our model aims to make expressive speech synthesis easily accessible through the use of synthetic data distillation, the ability also raises some authenticity and fairness concerns.

The primary ethical risk lies in the capacity to produce voices that are emotionally conditioned and realistic, which can be utilized as deepfake technology. We deploy two technical safeguards to address this problem. First, all demonstration audio is explicitly labeled as synthetic. Second, we integrate an inaudible watermarking mechanism based on the Perth framework by Resemble-AI [25], embedded directly at the waveform level. This watermark is undetectable by human listeners, yet it allows for effective post-hoc detection and guarantees traceability.

Our training datasets (EmoVoice-DB) were generated via GPT-4o-based TTS, which is fully synthetic. This design decision bypasses the privacy risks of recording human speakers but shifts the fairness burden to the teacher model, whose underlying data provenance is opaque, and it may harbor latent demographic or affective biases. We combat this uncertainty with transparency: we will thoroughly document the data generation pipeline and clearly state that all samples are AI-generated. This context is necessary for interpreting any stylistic or emotional biases observed for future research.

Bias can still emerge even from synthetic data. When specific emotions are linked to male or female timbres, emotional depictions may inadvertently perpetuate gendered or cultural prejudices. Moreover, the model's understanding of emotional prosody, which it learned in English, may not apply to other languages because intensity and expression vary a lot from culture to culture. Our mitigation regimen involves balanced sampling across timbres and emotions, monitoring classification accuracy partitioned by speaker identity, and reporting all systematic discrepancies. This process aims to promote responsible development in emotion-controllable TTS.

9 Conclusion

We show that expressive control can be transferred to a compact TTS model without large datasets or heavy architectures. By distilling from a larger teacher model into a 0.5B phoneme-based decoder, we add emotional and paralinguistic control while running faster than real time. Synthetic data, paired with a simple masking strategy, is sufficient to learn expressive cues without harming intelligibility or efficiency.

Our model performs competitively with a system six times larger and remains easy to deploy on standard hardware. Expressive behavior is learned from data rather than special architectural components, suggesting that emotion can be treated as a transferable capability rather than a feature limited to large proprietary models.

Limitations such as inherited artifacts from synthetic data, codec constraints, discrete emotion labels, and limited speaker diversity point toward clear future steps: more diverse data, codec adaptation, continuous emotion control, and evaluation over longer dialogue.

Overall, our results show that synthetic distillation offers a practical path to small, accessible, and expressive TTS systems. This helps shift focus from scaling up toward building models that are both human-like and deployable on everyday hardware.

10 Writing Revision

To address the feedbacks: **Abstract:** No significant changes as we keep the sentence that is no appropiate for proposal. **Data:** We added license information about the models and datasets. **Model Architecture:** We added more techinical details on the model architecture, we want to communicate and educate how a codec based TTS model works to 413 students. We changed the model diagram show how training and inference works, and which parts are frozen. **Work Div:** Eric did more work.

References

[1] Huda Barakat, Oytun Turk, and Cenk Demiroglu. "Deep learning-based expressive speech synthesis: a systematic review of approaches, challenges, and resources". In: *EURASIP Journal on Audio, Speech, and Music Processing* 2024.1 (2024), p. 11.

[2] Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N Chang, Sungbok Lee, and Shrikanth S Narayanan. "IEMOCAP: Interactive emotional dyadic motion capture database". In: *Language resources and evaluation* 42.4 (2008), pp. 335–359.

[3] Julia Cambre, Jessica Colnago, Jim Maddock, Janice Tsai, and Jofish Kaye. "Choice of voices: A large-scale evaluation of text-to-speech voice quality for long-form content". In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 2020, pp. 1–13.

[4] Deok-Hyeon Cho, Hyung-Seok Oh, Seung-Bin Kim, and Seong-Whan Lee. "DiEmo-TTS: Disentangled Emotion Representations via Self-Supervised Distillation for Cross-Speaker Emotion Transfer in Text-to-Speech". In: *arXiv preprint arXiv:2505.19687* (2025).

[5] Heejin Choi, Jae-Sung Bae, Joun Yeop Lee, Seongkyu Mun, Jihwan Lee, Hoon-Young Cho, and Chanwoo Kim. "Mels-Tts: Multi-Emotion Multi-Lingual Multi-Speaker Text-To-Speech System Via Disentangled Style Tokens". In: *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2024, pp. 12682–12686.

[6] Yu-An Chung, Yu Zhang, Wei Han, Chung-Cheng Chiu, James Qin, Ruoming Pang, and Yonghui Wu. *W2v-BERT: Combining Contrastive Learning and Masked Language Modeling for Self-Supervised Speech Pre-Training*. 2021. arXiv: `2108.06209 [cs.LG]`.

[7] Tri Dao. *FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning*. 2023. arXiv: `2307.08691 [cs.LG]`.

[8] Reece H. Dunn. *eSpeak NG Text-to-Speech*. Version 1.51.0. [Software]. 2024.

[9] Thierry Dutoit. *An introduction to text-to-speech synthesis*. Vol. 3. Springer Science & Business Media, 1997.

[10] Yiwei Guo, Chenpeng Du, Xie Chen, and Kai Yu. "Emodiff: Intensity controllable emotional text-to-speech with soft-label guidance". In: *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2023, pp. 1–5.

[11] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. "Distilling the knowledge in a neural network". In: *arXiv preprint arXiv:1503.02531* (2015).

[12] Rongjie Huang, Yi Ren, Jinglin Liu, Chenye Cui, and Zhou Zhao. "Generspeech: Towards style transfer for generalizable out-of-domain text-to-speech". In: *Advances in neural information processing systems* 35 (2022), pp. 10970–10983.

[13] Chae-Bin Im, Sang-Hoon Lee, Seung-Bin Kim, and Seong-Whan Lee. "EMOQ-TTS: Emotion Intensity Quantization for Fine-Grained Controllable Emotional Text-to-Speech". In: *ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2022, pp. 6317–6321.

[14] Sho Inoue, Kun Zhou, Shuai Wang, and Haizhou Li. "Hierarchical emotion prediction and control in text-to-speech synthesis". In: *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2024, pp. 10601–10605.

[15] Harry Julian, Rachel Beeson, Lohith Konathala, Johanna Ulin, and Jiameng Gao. "Finite Scalar Quantization Enables Redundant and Transmission-Robust Neural Audio Compression at Low Bit-rates". In: *arXiv preprint arXiv:2509.09550* (2025).

[16] Jaehyeon Kim, Jungil Kong, and Juhee Son. "Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech". In: *International Conference on Machine Learning*. PMLR. 2021, pp. 5530–5540.

[17] Tao Li, Shan Yang, Liumeng Xue, and Lei Xie. "Controllable emotion transfer for end-to-end speech synthesis". In: *2021 12th International Symposium on Chinese Spoken Language Processing (ISCSLP)*. IEEE. 2021, pp. 1–5.

[18] Shixiong Liang, Ruohua Zhou, and Qingsheng Yuan. "ECE-TTS: A Zero-Shot Emotion Text-to-Speech Model with Simplified and Precise Control". In: *Applied Sciences* 15.9 (2025), p. 5108.

[19] Steven R Livingstone and Frank A Russo. "The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS): A dynamic, multimodal set of facial and vocal expressions in North American English". In: *PloS one* 13.5 (2018), e0196391.

[20] Arian Dehghani Nazdridoy Arshiaesll. *Kokoro TTS*. 2025.

[21] Qwen, : An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. *Qwen2.5 Technical Report*. 2025. arXiv: `2412.15115 [cs.CL]`.

[22] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. 2024. arXiv: `2305.18290 [cs.LG]`.

[23] Yi Ren, Chenxu Hu, Xu Tan, Tao Qin, Sheng Zhao, Zhou Zhao, and Tie-Yan Liu. "Fastspeech 2: Fast and high-quality end-to-end text to speech". In: *arXiv preprint arXiv:2006.04558* (2020).

[24] Maya Research. *Maya1: Open Source Voice AI with Emotional Intelligence*. `https://huggingface.co/maya-research/maya1`. 2025.

[25] Resemble AI. *Perth: Audio Watermarking for Generative Speech Models*. `https://github.com/resemble-ai/perth`. GitHub repository, Accessed: November 2025. 2024.

[26] Jonathan Shen, Ruoming Pang, Ron J Weiss, Mike Schuster, Navdeep Jaitly, Zongheng Yang, Zhifeng Chen, Yu Zhang, Yuxuan Wang, Rj Skerrv-Ryan, et al. "Natural tts synthesis by conditioning wavenet on mel spectrogram predictions". In: *2018 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE. 2018, pp. 4779–4783.

[27] Hubert Siuzdak, Florian Grötschla, and Luca A. Lanzendörfer. *SNAC: Multi-Scale Neural Audio Codec*. 2024. arXiv: `2410.14411 [cs.SD]`.

[28] RJ Skerry-Ryan, Eric Battenberg, Ying Xiao, Yuxuan Wang, Daisy Stanton, Joel Shor, Ron Weiss, Rob Clark, and Rif A Saurous. "Towards end-to-end prosody transfer for expressive speech synthesis with tacotron". In: *international conference on machine learning*. PMLR. 2018, pp. 4693–4702.

[29] Haobin Tang, Xulong Zhang, Jianzong Wang, Ning Cheng, and Jing Xiao. "QI-TTS: Questioning Intonation Control for Emotional Speech Synthesis". In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–5.

[30] Noé Tits, Kevin El Haddad, and Thierry Dutoit. "Exploring transfer learning for low resource emotional tts". In: *Proceedings of SAI Intelligent Systems Conference*. Springer. 2019, pp. 52–60.

[31] Se-Yun Um, Sangshin Oh, Kyungguen Byun, Inseon Jang, ChungHyun Ahn, and Hong-Goo Kang. "Emotional speech synthesis with rich and granularized control". In: *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2020, pp. 7254–7258.

[32] Praveen Srinivasa Varadhan, Sherry Thomas, Suvrat Bhooshan, Mitesh M Khapra, et al. "The State Of TTS: A Case Study with Human Fooling Rates". In: *arXiv preprint arXiv:2508.04179* (2025).

[33] Tongzhou Wang, Jun-Yan Zhu, Antonio Torralba, and Alexei A Efros. "Dataset distillation". In: *arXiv preprint arXiv:1811.10959* (2018).

[34] Yuxuan Wang, Daisy Stanton, Yu Zhang, RJ-Skerry Ryan, Eric Battenberg, Joel Shor, Ying Xiao, Ye Jia, Fei Ren, and Rif A Saurous. "Style tokens: Unsupervised style modeling, control and transfer in end-to-end speech synthesis". In: *International conference on machine learning*. PMLR. 2018, pp. 5180–5189.

[35] Detai Xin, Xu Tan, Shinnosuke Takamichi, and Hiroshi Saruwatari. *BigCodec: Pushing the Limits of Low-Bitrate Neural Speech Codec*. 2024. arXiv: `2409.05377 [eess.AS]`.

[36] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jianxin Yang, Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai, Sinan Tan, Tianhang Zhu, Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Xuejing Liu, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yuqiong Liu,

Zeyu Cui, Zhenru Zhang, Zhifang Guo, and Zhihao Fan. *Qwen2 Technical Report*. 2024. arXiv: 2407.10671 [cs.CL].

[37] Guanrou Yang, Chen Yang, Qian Chen, Ziyang Ma, Wenxi Chen, Wen Wang, Tianrui Wang, Yifan Yang, Zhikang Niu, Wenrui Liu, Fan Yu, Zhihao Du, Zhifu Gao, ShiLiang Zhang, and Xie Chen. *EmoVoice: LLM-based Emotional Text-To-Speech Model with Freestyle Text Prompting*. 2025. arXiv: 2504.12867 [eess.AS].

[38] Neil Zeghidour, Alejandro Luebs, Ahmed Omran, Jan Skoglund, and Marco Tagliasacchi. "SoundStream: An End-to-End Neural Audio Codec". In: *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 30 (2022), pp. 495–507.